

Predicting House Prices

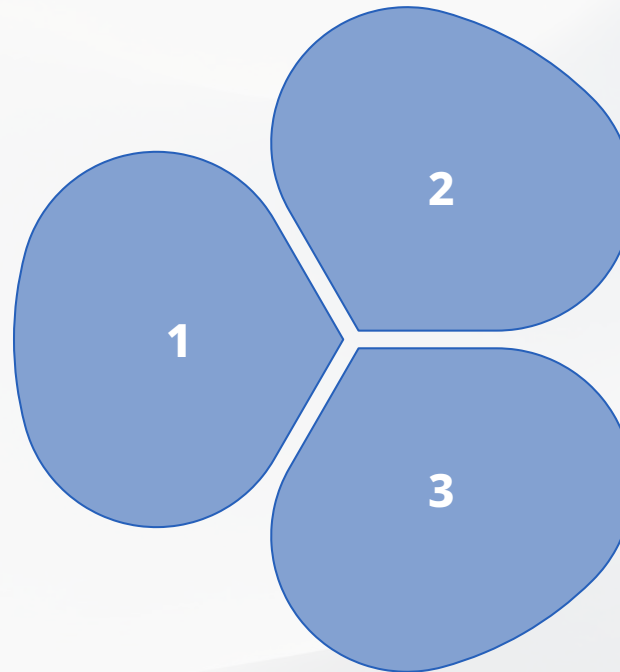
Machine Learning for Real Estate Price Prediction using Linear Regression

Tools Used: Python, Pandas, Scikit-learn, Seaborn, Matplotlib, Jupyter Notebook¹

Project Objective

Business Problem

Real estate prices vary based on multiple property characteristics, making accurate valuation difficult without data-driven tools



Objective

Build a Linear Regression model to estimate house prices from key property features using historical data

Expected Outcome

Accurate price prediction with interpretable feature importance for buyers, sellers, and agencies



Dataset Overview

Housing Dataset Features

Loaded with Pandas — contains both numerical and categorical variables, so preprocessing is required.



Price

Target variable — the value to be predicted



Area

House size in square feet



Bedrooms & Bathrooms

Number of bedrooms and bathrooms



Stories & Parking

Number of floors and parking spaces



Location & Age

Neighborhood category and property age

	A	B	C	D	E	F	
1	postcode	bedrooms	bathrooms	parkingSpaces	propertyType	price	
2	3227	3	2	2	house	\$1,255,000	
3	3226	4	2	2	house	\$725,000	
4	3064	4	2	2	house	\$670,000	
5	3215	3	2	1	house	\$600,000	
6	3218	3	1	2	house	\$520,000	
7	3234	3	2	3	house	\$515,000	
8	3212	3	1	1	house	\$460,000	
9	3806	5	4	3	house	\$1,935,000	
10	3243	3	1	2	farmlet	\$1,500,000	
11	3168	5	2	2	house	\$1,400,000	
12	3941	3	3	3	house	\$1,300,000	
13	3138	5	2	2	house	\$990,000	
14	3429	4	3	5	house	\$920,000	
15	3136	4	2	2	house	\$885,000	
16	3072	3	1	1	house	\$855,000	
17	3023	5	3	2	house	\$855,000	
18	3821	4	2	6	house	\$835,000	
19	3034	3	2	1	house	\$750,000	

Exploratory Data Analysis



Distribution of House Prices

Most houses belong to the medium price range. Distribution is right-skewed with luxury homes appearing as high-value outliers



Key Insight

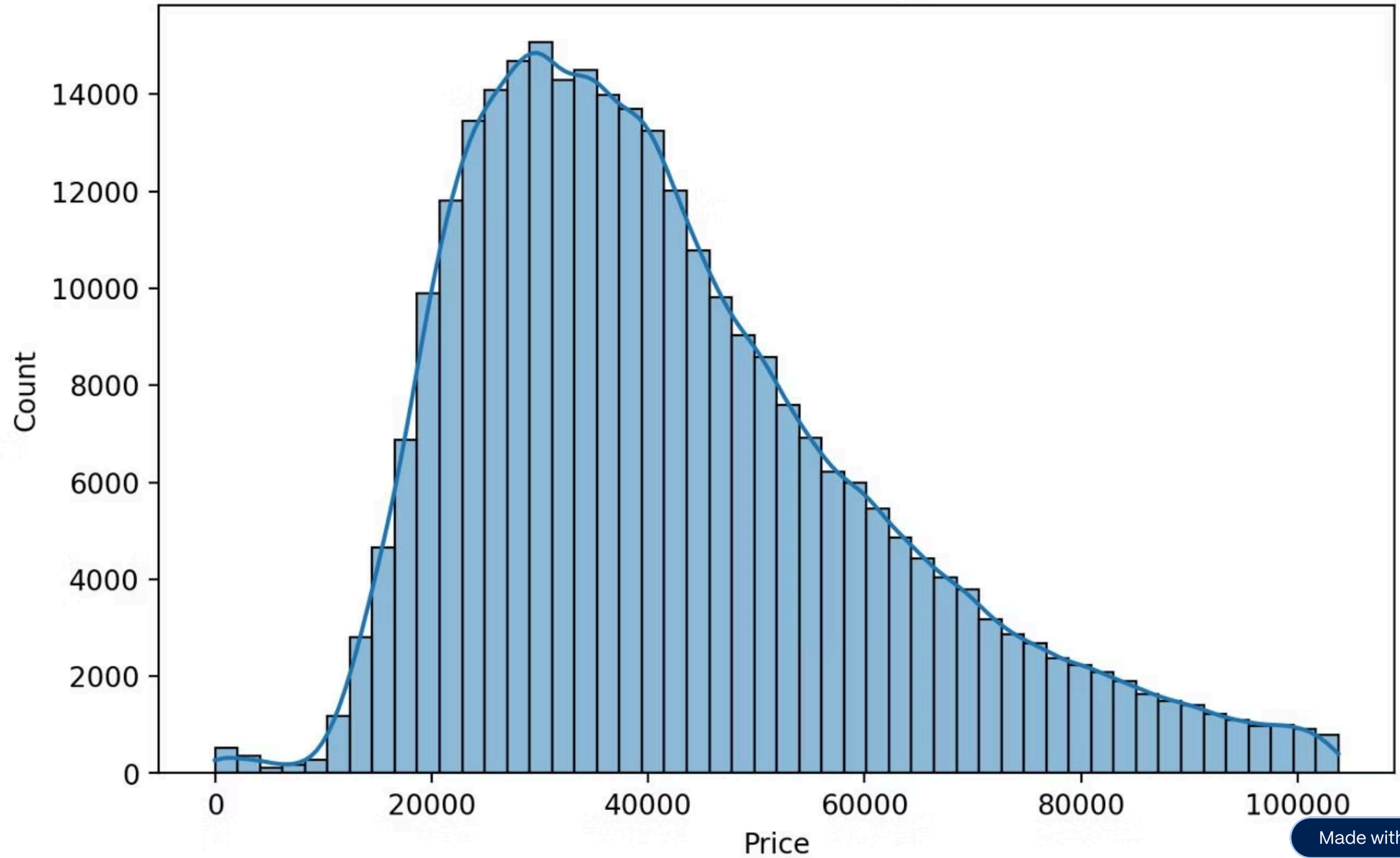
Understanding the target variable helps identify unusual observations before model training



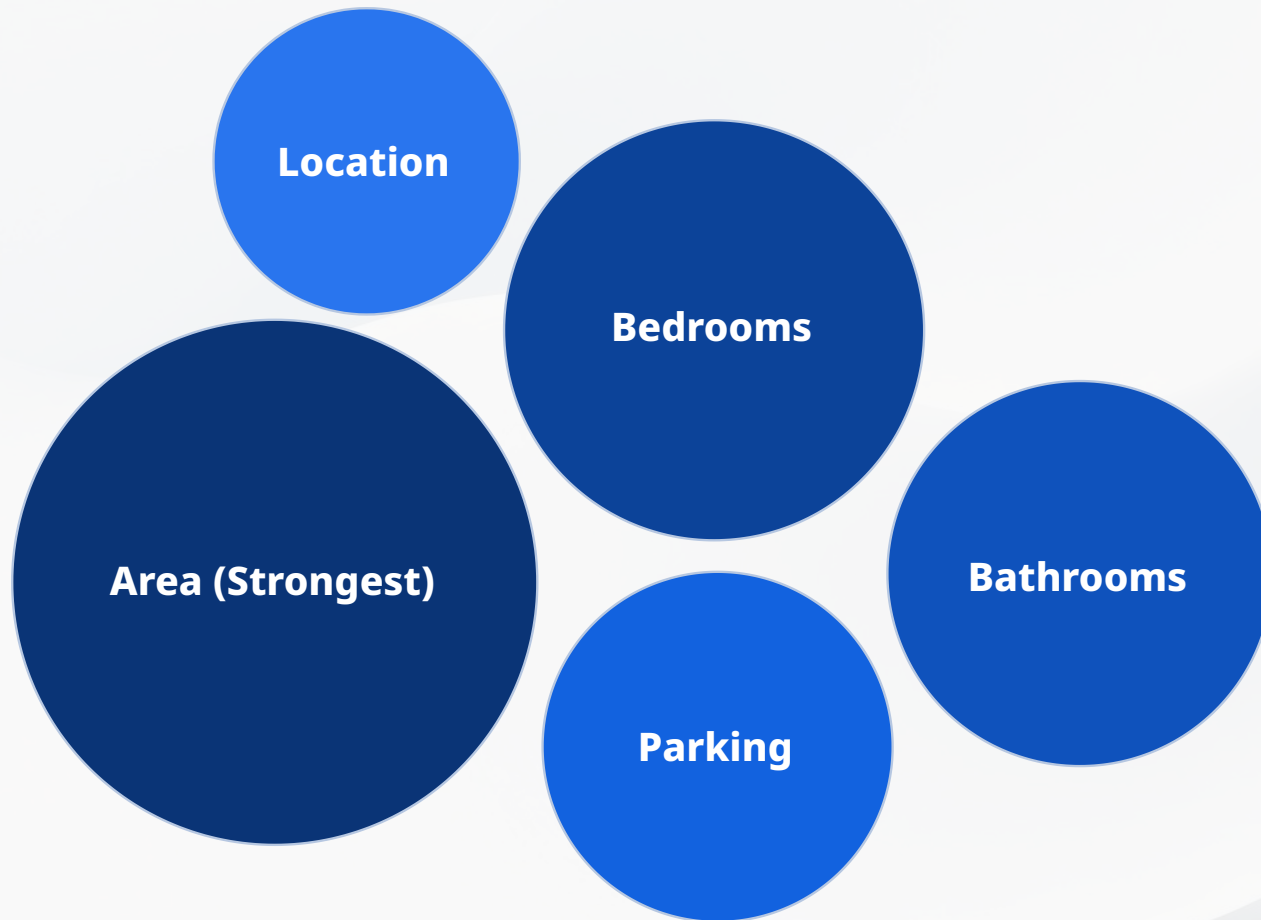
Feature Relationships

Visualize correlations between area, bedrooms, bathrooms, and price using pair plots and heatmaps

Distribution of Housing Prices



Feature Selection



Selection Criteria

Does the feature have a measurable impact on price?

Is the feature consistent and available in real data?

Does it align with known real estate valuation logic?

Is it statistically correlated with the target variable?

Feature selection combines business domain knowledge with statistical relevance.



Data Preprocessing

01

Handle Missing Values

Inspect and impute or remove incomplete records to ensure a clean dataset

03

Prepare Numerical Features

Verify data types and scale numerical inputs for modeling compatibility

02

Encode Categorical Variables

Convert the Location column using One-Hot Encoding — Urban, Rural, Suburban become binary columns

04

Final Dataset Validation

Confirm no nulls remain and all features are in the correct format before model training

One-Hot Encoding

										
Embarked	A	B	A	B	C	A	B	C	A	B
Is_A	1	0	1	0	0	1	0	0	1	0
Is_B	0	1	0	1	0	0	1	0	0	1
Is_C	0	0	0	0	1	0	0	1	0	0

Label	Is_A	Is_B	Is_C
A	1	0	0
B	0	1	0
C	0	0	1

Correlation Heatmap

Relationship Between Features

A heatmap of `df.corr()` reveals which features most strongly influence house prices.

Area → Price

Strongest positive correlation (0.54)
— the single most influential predictor

Bathrooms → Price

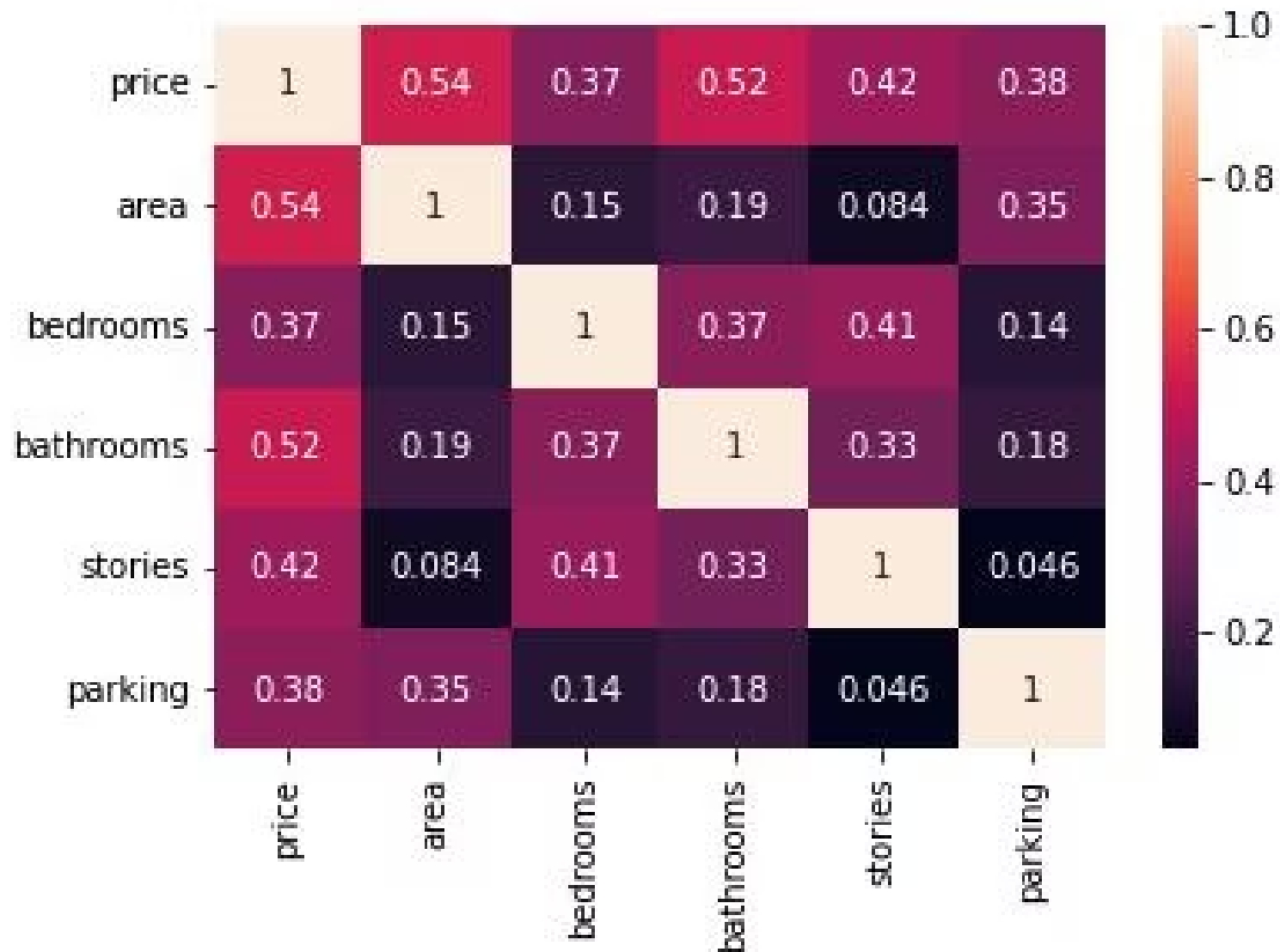
Second strongest relationship (0.52)
— adds measurable property value

Bedrooms → Price

Moderate positive correlation (0.37)
— indicates living capacity

```
sns.heatmap(df.corr(), annot=True)
```

<AxesSubplot:>



Linear Regression Model

Model Building Process

Using Scikit-learn to train a Linear Regression model on the preprocessed housing dataset.



Split Dataset

80% training / 20% test split to evaluate generalization performance



Train Model

Fit Linear Regression on training data using
`LinearRegression().fit()`



Predict Prices

Generate predictions on the test set and compare against actual values

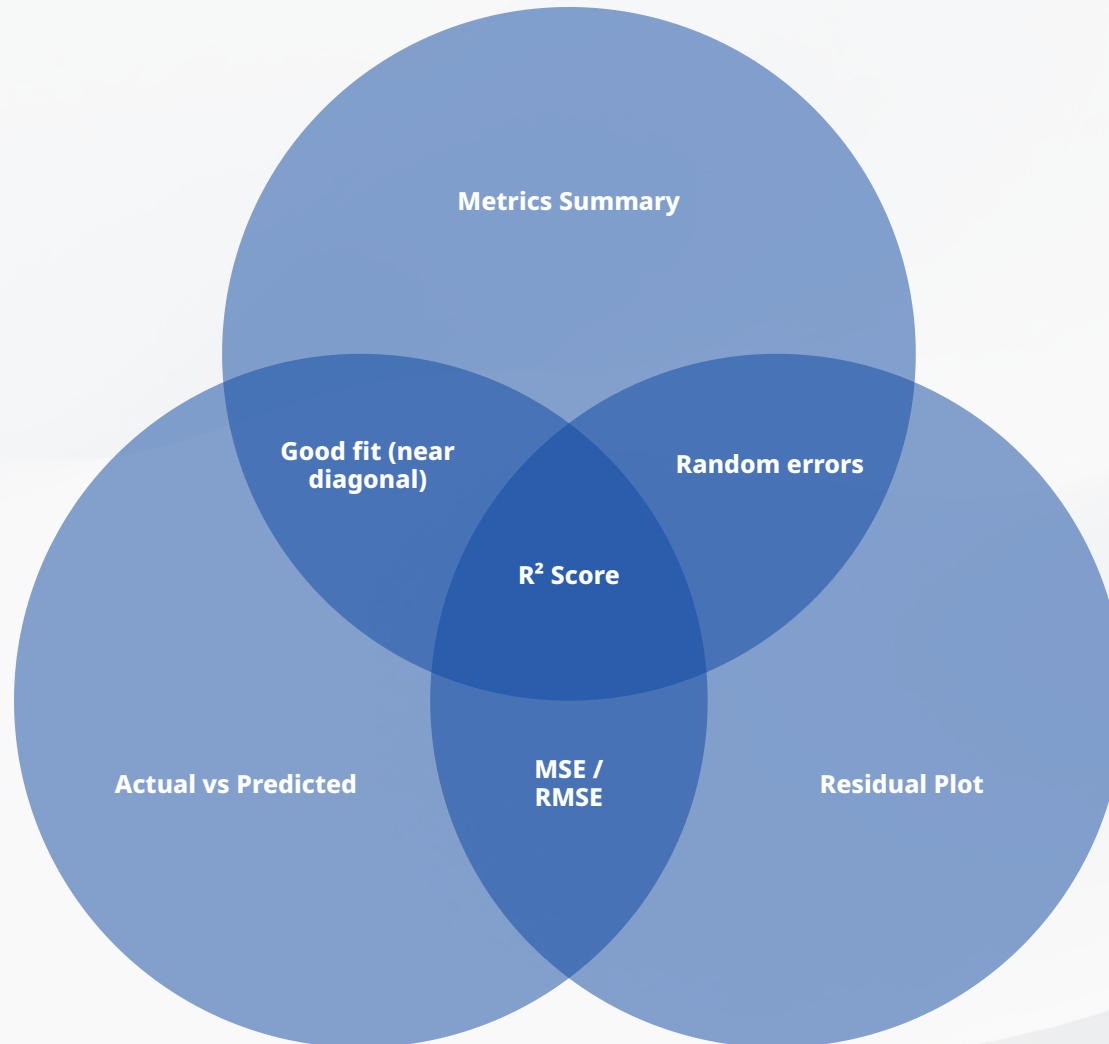


Why Linear Regression?

Predicts continuous numerical values and provides easily interpretable coefficients

Linear Regression maps feature inputs to a continuous price output — making it ideal for interpretable real estate valuation.

Model Evaluation



Evaluation Metrics

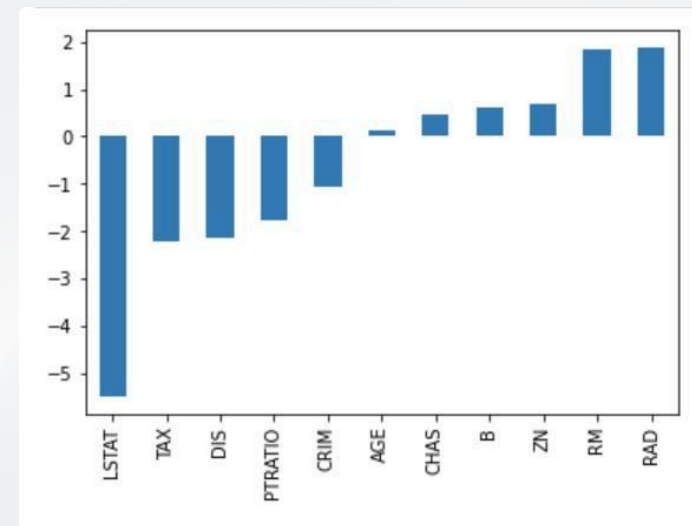
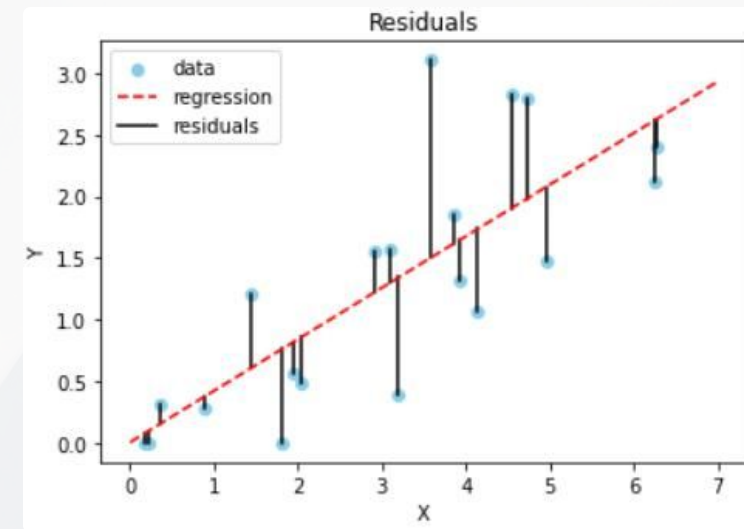
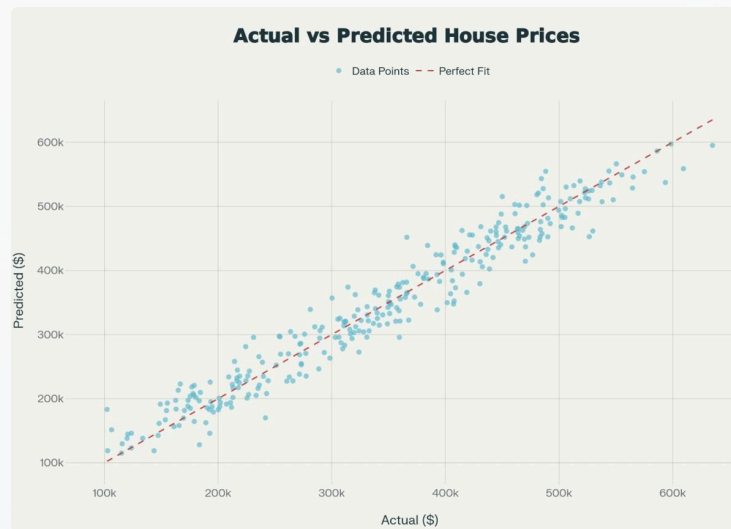
MSE — Measures average squared prediction error

RMSE — Average error in the same unit as price

R^2 Score — Overall model accuracy and fit quality

Residual Plot — Validates error distribution assumptions

Points close to the diagonal in the scatter plot indicate better predictions from the model.



Conclusion & Business Recommendations

Key Findings



Area is Dominant

House area has the highest positive impact on predicted price across all features.



Bathrooms & Parking Add Value

These features consistently increase property valuation in the model.



Prioritise Location & Area

Use these as primary factors during property valuation and pricing decisions.



Retrain Periodically

Update the model regularly with fresh market data to maintain prediction accuracy.

Linear Regression successfully predicts housing prices and provides interpretable business insights for data-driven real estate decisions.